

BMJ Open Multilayer low-stretch bandage combined with conventional rehabilitation to promote functional recovery after knee surgery: study protocol for a prospective, single-blind, randomised controlled trial

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ABSTRACT

Introduction Patients undergoing anterior cruciate ligament reconstruction (ACLR) and patellar fracture frequently present with substantial pain, swelling and inflammation. These pathophysiological changes not only intensify postoperative pain but also compromise the intra-articular environment via mechanical compression and localised microcirculatory deficits, thereby impeding recovery. Conventional interventions for postoperative swelling, such as cryotherapy and physical agent modalities, are commonly applied yet exhibit limited efficacy. Multilayer low-stretch bandage (MLB), with its gradient compression properties, has recently demonstrated superior swelling control and functional restoration; however, existing research primarily assesses short-term outcomes and lacks a systematic analysis across the postoperative continuum—acute, subacute and chronic phases—as well as long-term follow-up. This study therefore employs a multi-temporal intervention design and extended follow-up to evaluate the effectiveness of MLB in managing swelling, modulating pain and enhancing long-term knee function and activities of daily living across different recovery stages. The findings aim to furnish high-level evidence for knee rehabilitation protocols, refine clinical practice and ultimately improve patient quality of life.

Methods and analysis This prospective, single-blind, randomised controlled trial will enrol 36 patients following ACLR and patellar fracture. Participants will be allocated at a 1:1 ratio through computerised randomisation to either an experimental group receiving MLB plus conventional rehabilitation, or a control group receiving conventional rehabilitation alone. The study spans 12 weeks, with follow-up assessments scheduled at baseline (T0), 4 weeks (T1), 8 weeks (T2) and 12 weeks (T3). Outcome measures comprise the Hospital for Special Surgery (HSS) Knee Rating Score, affected limb oedema regression rate, Visual Analogue Scale (VAS) pain score, knee range of motion, quadriceps and hamstring muscle strength, centre of pressure (COP) displacement parameters, proprioceptive testing, the 36-Item Short Form Health Survey (SF-36) and the Holden Walking Function

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ A prospective, randomised controlled design facilitates a direct comparison of the combined intervention versus conventional rehabilitation.
- ⇒ Outcome assessment uses a comprehensive set of objective and patient-reported measures to evaluate multiple functional domains.
- ⇒ A predefined statistical analysis plan was finalised prior to data collection.
- ⇒ The single-centre design may limit the generalisability of the findings.
- ⇒ The open-label design carries a risk of performance and detection bias.

Classification. A subsequent 1-year follow-up will assess long-term efficacy and patient satisfaction, while adverse events are strictly monitored throughout the entire study period. Adherence to Consolidated Standards of Reporting Trials guidelines is maintained, and a single-blind design ensures that both participants and assessors remain unaware of group assignments. This trial aims to establish an evidence-based foundation for optimising postoperative rehabilitation in this patient population.

Ethics and dissemination Ethical approval for the study was obtained from the Medical Ethics Committee of Nanjing Drum Tower Hospital, the Affiliated Hospital of Nanjing University Medical School (2025-0141-01). The results of the study will be published in a peer-reviewed medical journal.

Trial registration number ClinicalTrials.gov
chiCTR2500100566. Registration date is 10 April 2025.

INTRODUCTION

Patients undergoing anterior cruciate ligament reconstruction (ACLR) or patellar fracture surgery frequently experience substantial pain, swelling and inflammation. These pathophysiological changes not only intensify postoperative pain but also impede

healing by compromising the intra-articular environment through mechanical compression and localised microcirculatory deficits.¹ As the primary clinical manifestation in the early phase, swelling markedly impairs the patient's capacity to perform daily activities.² It elevates pain levels and, more critically, compresses adjacent blood vessels and nerves, thereby restricting joint mobility.² This compression promotes the release of local inflammatory mediators, induces microcirculatory dysfunction and contributes to muscle atrophy.³ Ultimately, these alterations disrupt the normal tissue repair process.

Although existing treatments can alleviate swelling to some extent, their efficacy remains limited, particularly for prolonged swelling; this inadequacy directly impacts surgical outcomes and the patient's overall recovery process.⁴ Traditional Chinese medicine approaches such as acupuncture and cupping possess distinctive characteristics in managing postoperative swelling, yet they face challenges in standardisation and quantitative assessment. High-elasticity bandages are widely employed due to their immediate compression effect, although prolonged use may cause skin discomfort and restrict blood circulation.^{5,6} Cold therapy effectively reduces inflammation and exudation, whereas heat therapy promotes muscle relaxation and blood circulation, yet the benefit of either modality alone is limited.⁷ Physical factor therapy and manual therapy can improve local blood circulation and promote lymphatic reflux to a degree, but their effectiveness is highly variable across individuals and their application is complex.⁸ While drug therapy rapidly relieves symptoms, long-term administration is accompanied by side effects.

In recent years, multilayer low-stretch bandage (MLB) has attracted research interest for its ability to provide sustained, moderate pressure that promotes lymphatic return while minimising adverse effects on skin and circulation. Studies indicate that multilayer compression therapy effectively reduces early postoperative ankle oedema⁹ and significantly improves knee range of motion (ROM) and straight leg raise ability during the early postoperative phase following total knee arthroplasty (TKA).¹⁰ However, most existing research has emphasised short-term outcomes, lacking systematic evaluation of intervention effects across postoperative stages—acute, subacute and chronic—as well as long-term follow-up.¹¹ Given the notable advantages of MLB in oedema relief and its established role in preoperative rehabilitation, further investigation is warranted. This randomised controlled trial (RCT) will aim to evaluate the efficacy of MLB combined with conventional rehabilitation in improving long-term functional outcomes among patients who have undergone ACLR and patellar fracture surgery. Specifically, the study will investigate whether this combined intervention enhances swelling control and functional recovery across the acute, subacute and chronic postoperative phases. It will also examine the intervention's potential to accelerate patients' return to normal activities by optimising the rehabilitation pathway.⁴ In addition, this study will be

expected to provide a scientific basis for clinical decision-making, promote the personalisation and precision of postoperative rehabilitation programmes, and further improve the overall treatment outcomes and quality of life in patients with knee injuries.⁸ Through extended intervention implementation and follow-up assessments, this research will seek to establish an evidence-based foundation for personalising postoperative rehabilitation protocols in the management of knee injuries.

METHODS

Study design

This prospective, single-blind RCT will randomly assign participants—patients with postoperative swelling and dysfunction after ACLR or patellar fracture—in a 1:1 ratio to either the experimental or control group. The trial adheres to the Declaration of Helsinki and has been approved by the Ethics Committee of Nanjing Drum Tower Hospital, the Affiliated Hospital of Nanjing University Medical School. It also complies with the Standard Protocol Items: Recommendations for Interventional Trials 2010 Statement¹² and relevant reporting guidelines.¹³

Study setting and recruitment

This study will be conducted within the Department of Rehabilitation Medicine at Nanjing Drum Tower Hospital, the Affiliated Hospital of Nanjing University Medical School, a tertiary care facility in Nanjing, China. Participants will be consecutively recruited in collaboration with the hospital's Joint Surgery Department between 2025 and 2027. Recruitment will take place from the Department of Rehabilitation Medicine. Patients are first evaluated by a Physical Medicine and Rehabilitation specialist after being discharged from the Orthopaedic Surgery department 2–3 days postsurgery. Those exhibiting clinical signs of arthrofibrosis following ACLR and postoperative swelling after patellar fracture are then referred to a study physiotherapist. The physiotherapist subsequently assesses their eligibility against all predefined inclusion and exclusion criteria. Eligible patients who provide written informed consent are formally enrolled to receive the assigned intervention. They will be informed of their random assignment to a study group, that each group receives a different technique appropriate for their condition, and that the study aims to determine which technique yields the best outcomes. On agreeing to participate, they will sign an informed consent form, after which the same recruiting physiotherapist will perform a baseline assessment.

Patient and public involvement (PPI) statement

This study established a patient advisory panel of five members who were engaged throughout the research design and implementation. During protocol development, the panel helped refine research questions, optimise outcome measures and assess the acceptability of the

study design. In the execution phase, two patient representatives collaborated to improve recruitment materials, while one public contributor served as a co-investigator in data collection and interpretation. The research findings will be disseminated through community seminars and patient-friendly reports, with all academic outputs reviewed by patient representatives to ensure clarity and accessibility.

Informed consent

Written informed consent will be obtained from all participants prior to the initiation of any research procedures. The consent process includes a detailed explanation of the study's objectives, procedures, potential risks and benefits. Participants will be explicitly informed of their right to withdraw from the study at any time without penalty. The research team will ensure that all participants fully comprehend this information and will encourage their continued participation through both the treatment and follow-up phases of the study.

Randomisation, allocation concealment and blinding

Participants will be randomly assigned to study groups using a computer-generated sequence. Allocation will be concealed with sequentially numbered, opaque, sealed envelopes. An independent research therapist, who will not be involved in recruitment or outcome assessment, will manage the envelopes and administer the interventions. This procedure will ensure that participants remain blinded to their group assignment for the study's duration.

Criteria for inclusion and exclusion

Inclusion/exclusion criteria are listed in [table 1](#).

Sample size calculation

Sample size estimation was conducted using PASS software, version 2024 (NCSS, Kaysville, Utah, USA), employing an independent samples t-test. Drawing from pilot data in previous literature that reported postintervention values for the primary outcome (Control:

80.7±5.2 vs Experimental: 85.7±3.9),¹⁴ parameters were set with $\alpha=0.05$ and a power of 0.8. Initial calculations determined a minimum need for 15 participants in each group ($n_1=n_2=15$) with a 1:1 allocation ratio. Considering a potential 20% dropout rate, the final recruitment goal was adjusted to 19 subjects per group (total $n=38$). Although our a priori sample size calculation, based on the best available preliminary data, indicated that 38 participants would provide 80% power, we have opted for a more robust recruitment strategy. To improve the statistical reliability and generalisability of our findings, we will proactively enrol additional participants beyond this initial target. Participants will be consecutively recruited from the Department of Joint Surgery at Nanjing Drum Tower Hospital from 2025 to 2027. All surgical procedures will be performed by the same surgical team under standardised intraoperative conditions to ensure consistency. Strict inclusion and exclusion criteria will be implemented to minimise confounding variables.

Flow

The physiotherapist responsible for recruitment will conduct all evaluations while remaining blinded to group assignments. Initial recruitment will strictly adhere to the predefined inclusion and exclusion criteria outlined in [table 1](#). Eligible participants must provide written informed consent before the study commences. A baseline assessment will collect general patient information, including name, age, gender, height, weight, affected knee side, surgical date, injury date and injury type. Participants will then be randomly allocated to either the experimental or control group. Four assessment time points—baseline (T0), and at 4 weeks (T1), 8 weeks (T2), and 12 weeks (T3) postintervention—will collect outcome data encompassing the HSS Knee Rating Score, affected limb oedema reduction rate, VAS pain score, knee ROM, quadriceps/hamstrings muscle strength, COP displacement parameters, proprioceptive test results, SF-36 scores, and Holden Walking Function Classification grades. The study will also document special circumstances, such as

Table 1 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
▶ Patients diagnosed with ACLR and postoperative swelling, pain and loss of function after patellar fracture ▶ Limb circumference greater than 2 cm or volume increase of 10% or more compared with the contralateral side ▶ Age 18–60 years ▶ No other rehabilitation treatment ▶ Patients who voluntarily signed the informed consent and participated in this study	▶ Serious complications (eg, infection, deep Vein thrombosis, neurovascular injury) ▶ Other systemic diseases (severe systemic diseases of the heart, lungs, liver, kidneys, etc) ▶ Patients with cognitive impairments or psychiatric disorders that prevent them from cooperating with the completion of all treatment and assessment indicators required for the study should be excluded ▶ Patients with allergies to MLB materials should be excluded ▶ Infected or badly broken limbs ▶ Use of any medication that affects body fluids and electrolytes ▶ Other conditions that the researcher considers inappropriate for participation in this study

ACLR, anterior cruciate ligament reconstruction; MLB, multilayer low-stretch bandage.

protocol deviations, along with patient dropout rates during the intervention period. Follow-up visits at 3, 6 and 12 months postintervention will gather additional patient data. The complete experimental procedure is detailed in [figure 1](#).

Interventions

The rehabilitation programme integrated the 2021 AAOS Guidelines¹⁵ for Postoperative Orthopaedic Rehabilitation with the Guidelines for Postoperative Rehabilitation Treatment of ACLR and Patellar Fractures from Sports Injury Science Prevention, Treatment and Rehabilitation¹⁶ and the Postoperative Rehabilitation Programme for ACLR and Patellar Fractures developed by the Department of Rehabilitation Medicine at the Drum Tower Hospital, Affiliated to the Medical College of Nanjing University. All participants received standardised perioperative rehabilitation education before the intervention began. This curriculum, derived from evidence-based guidelines, covered postoperative rehabilitation protocols, standardised cryotherapy, assistive device use, multimodal analgesia, activity modification precautions and the recognition and management of complications. A uniformly trained therapist team delivered all educational content to ensure intervention consistency. For postoperative oedema management, a standardised protocol included ankle pump exercises (20 repetitions per set, 10 sets daily) and limb elevation in a supine position with the foot raised 20 cm above the cardiac level for 30 min per session, three times daily. Regulated cryotherapy was applied to address postoperative pain and localised skin temperature elevation, with sessions lasting 20 min and separated by intervals of at least 2 hours. Throughout the trial, any deviations from the prescribed protocol or use of concomitant therapies were meticulously documented. All rehabilitation procedures followed predetermined standardised protocols to maintain intervention consistency and comparability.

Participants will be randomly assigned to either Group A or Group B. Group A will receive conventional rehabilitation intervention, whereas Group B will undergo the same conventional rehabilitation protocol supplemented with MLB therapy. The MLB intervention will use a standardised three-layer bandage system (Lohmann and Rauscher International GmbH & Co. KG; [figure 2](#)), applied according to the following procedure. A Tg tubular bandage will form the first layer to protect the skin and absorb moisture, applied smoothly from the dorsum of the foot upward without folds or indentations. The second layer will consist of Rosidal Soft protective padding to reduce pressure-related skin damage and prevent friction or shearing, applied in a distal-to-proximal spiral pattern with a 1/3 overlap. Rosidal K E low-elastic bandage will constitute the third layer, applied from distal to proximal using a standard figure-of-eight wrapping technique, with specific angular crosses at the joints. A Mollelast bandage will also be used selectively for specialised toe management in the presence of pedal oedema. The starting point

for bandage application will be individualised according to the extent of oedema: beginning at the toes for toe swelling, at the dorsum of the foot for foot swelling, and at the calf level for isolated calf swelling. All bandage applications will extend to the mid-thigh. The target pressure of the bandage system will be maintained between 20 and 50 mm Hg, adjusted based on patient tolerance. Participants will wear the bandage system for 20–22 hours daily to ensure therapeutic efficacy while allowing time for skin care, personal hygiene and conventional rehabilitation. Daily assessments will monitor skin condition and guide any necessary treatment adjustments. [Figure 3](#) illustrates the clinical application of MLB for knee joint swelling. Routine rehabilitation sessions will be conducted three times per week, each lasting 60 min, over a 12-week period. A detailed description of the intervention programme is provided in [table 2](#).

Data collection and analysis

Date collection

The same therapist conducted data collection both prior to and following each indicator assessment, whereas outcome measurements were consistently tracked and recorded by the corresponding specialists over the entire study period.

Baseline data: information on patient demographics and pretreatment conditions was collected. This included general patient information, the HSS, rate of oedema reduction in the affected limb, VAS, knee ROM, muscle strength testing (quadriceps/hamstrings), COP displacement parameters, proprioceptive testing, the SF-36 and Holden Walking Function Classification.

Follow-up data: patient prognostic variables were collected by regularly assessing outcome indicators at the four visit points and at follow-up visits 3, 6 and 12 months after the end of the intervention, as shown in [table 3](#).

Outcome measurements

[Table 3](#) lists the outcome variables, their respective assessment time points, and the corresponding measurement tools. The primary outcome is the HSS score, and secondary outcomes include the rate of oedema resolution in the affected limb, VAS, knee ROM, muscle strength testing (quadriceps/hamstrings), COP displacement parameters, proprioceptive testing, SF-36 and Holden Walking Functional Classification. A physical therapist will perform all assessments and oversee participant recruitment. To minimise error, the physical therapist will not be informed of each participant's group assignment during the assessment.

Hospital for special surgery knee score

The HSS is a standardised instrument for evaluating knee function and surgical results. It comprises seven questions, with six items—pain, function, ROM, muscle strength, flexion deformity and stability—contributing to the score, while one item encompassing the use of walking aids, extension lag and valgus deformity is subtracted.

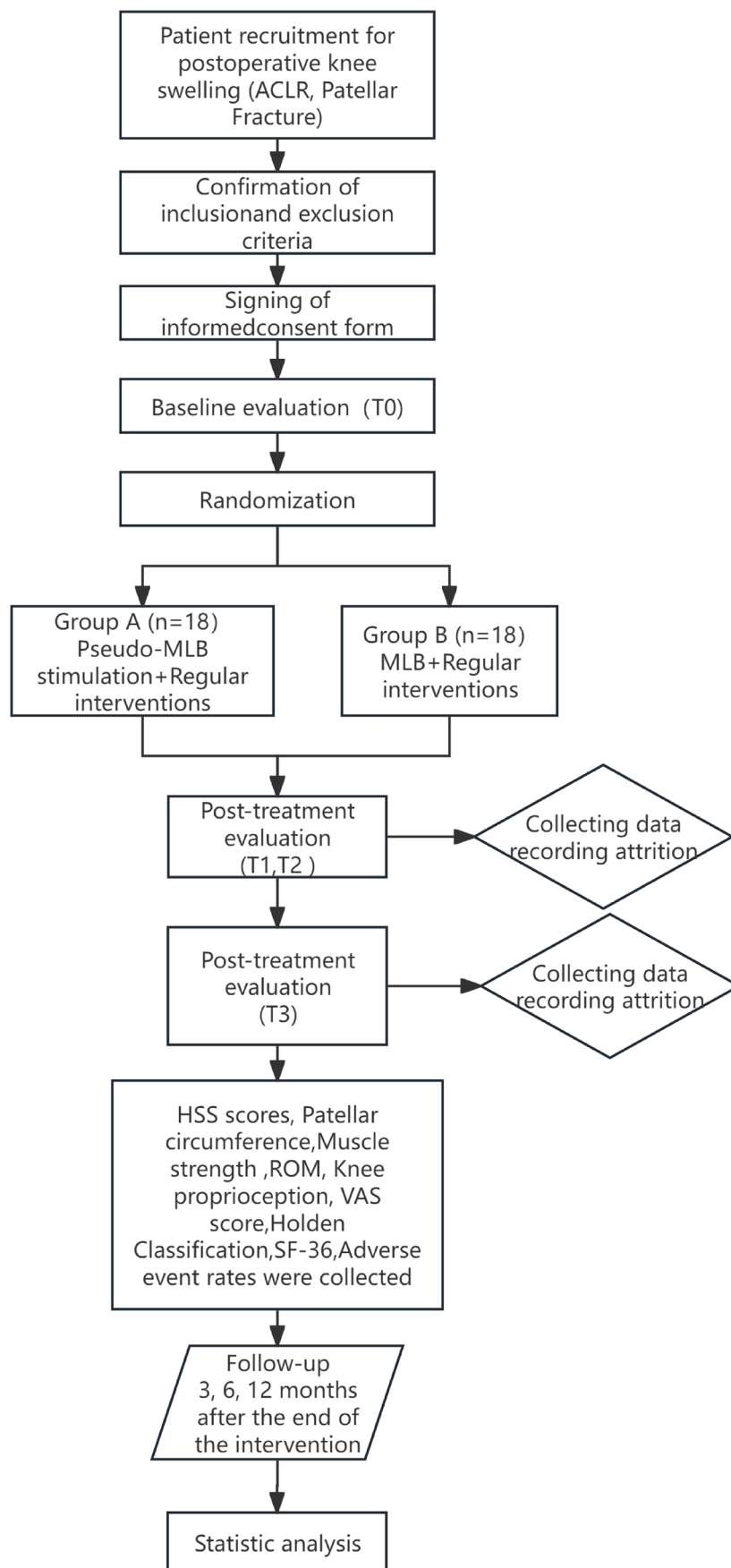


Figure 1 Flowchart of the experiment: flow of the participants throughout the study. ACLR, anterior cruciate ligament reconstruction; MLB, multilayer low-stretch bandage; ROM, range of motion; SF-36, 36-Item Short Form Health Survey; VAS, visual analogue scale.



Figure 2 Multilayer low-stretch bandage components.

The total score is 100 points, where a score of ≤ 59 denotes poor function, 60–69 fair, 70–84 good and ≥ 85 excellent; higher scores reflect superior knee function.¹⁷ Trained physiotherapists, blinded to group allocation, will administer the HSS at T0, T1, T2 and T3. Standardised protocols will be followed, including pain assessment through structured interviews using a validated six-level descriptive scale, ROM measurement with a universal goniometer in the supine position, and muscle strength evaluation via manual muscle testing (MMT) according to established guidelines.

Patellar circumference

Knee joint swelling will be objectively quantified by circumferential measurements taken at the mid-patellar level. With the participant supine and the knee in a relaxed, extended position, the patellar midpoint will be palpated and marked on the affected knee. A flexible, non-stretch tape measure will then be applied

horizontally at this mark, maintaining consistent skin contact without compressing the soft tissues. The circumference will be recorded to the nearest millimetre. The same procedure will be performed on the contralateral knee to obtain a baseline reference value. All measurements will be conducted by a single trained examiner to ensure consistency, with duplicate measurements taken at each assessment time point for reliability. The circumference of the contralateral patella was measured in the same way as a reference.¹⁸

VAS score

VAS for pain assessment will be administered according to a standardised protocol. The assessment form features a 10 cm horizontal line anchored by 'no pain' at the 0 mm endpoint and 'worst imaginable pain' at the 100 mm endpoint.¹⁹ A trained researcher will instruct each participant verbally: 'Please mark a vertical line on this scale to indicate your current knee pain intensity during rest. The left end represents no pain, while the right end represents the most severe pain you can imagine'. Participants then mark the scale independently, without researcher intervention. The researcher measures the distance from the left endpoint to the mark with a ruler and records the value in millimetres. This assessment is repeated at each designated evaluation time point using the same standardised procedure to maintain consistency.

Knee ROM

Knee ROM serves as a useful indicator of functional recovery and is measured with a goniometer.²⁰ Measurements are taken with the subject in a supine position, arms resting naturally at the sides, and thighs exposed to locate the greater trochanter. The goniometer's centre fulcrum is positioned over the lateral epicondyle of the femur. The proximal arm is



Figure 3 Clinical application of a multilayer low-stretch bandage.

Table 2 Specific interventions for each group

	Group A (control group) routine rehabilitation		Group B (experimental group) MLB+routine rehabilitation
	Postoperative ACLR	Postoperative patella fracture	
Phase I (0–2 weeks) Baseline data	1. Protecting the postoperative knee 2. Control of inflammation (ankle pump, elevation of affected limb) 3. Restoration of quadriceps contraction (isometric contraction, straight leg raise, quadriceps resisted straight leg raise) 4. Mobility exercises (extension exercises, passive knee flexion exercises) 5. Minimising the impact of fixation	1. Restore ROM 2. Prevention and control of pain and swelling (ice, elevation of the affected limb, ankle pump) 3. Mobility training (loosening patella, passive knee movement, adjacent joint mobility training) 4. Maintenance of muscle strength (knee joint muscle strength training, hip joint muscle strength training) 5. Prevention of lower extremity venous thrombosis	1. Addition of MLB: MLB is the use of multi-layer low elastic bandages to apply pressure for 22 hour/day 2. The rest of the interventions were the same as those in the stages of routine rehabilitation
Phase II (2–6 weeks)	1. Unrestricted extension 2. Controlled standing on one leg 3. Pain-free stepping and micro squatting 4. Good quadriceps control 5. Gradual release from crutches and braces 6. ROM training	1. To further restore joint ROM 2. Prevention and control of pain and swelling 3. Muscle strength exercises 4. Increases in balance training 5. Progressive weight-bearing exercises	
Phase III (6–12 weeks)	1. Normalise gait 2. Increased quadriceps strength 3. Improved neuromuscular control of the ankle and hip 4. Improved balance and proprioception	1. Restoration of residual strength deficit 2. Progressive restoration of motor function	

Note: avoid significant knee joint pain and swelling during all of the above exercises.

ACLR, anterior cruciate ligament reconstruction; MLB, multilayer low-stretch bandage; ROM, range of motion.

aligned with the lateral midline of the femur, using the greater trochanter as a reference point, while the distal arm is aligned with the lateral midline of the fibula, referenced from the lateral malleolus and the fibular head.²¹ The examiner, seated on a stool,

aligned the goniometer at eye level for reading. The patient's knee was passively flexed to its maximum angle for each measurement, and this procedure was repeated three times. The mean value of these measurements was recorded.

Table 3 Outcome variables

Outcome variables	Evaluation period				Measuring instrument
	Baseline	T1	T2	T3	
Strength	×	×	×	×	Manual muscle testing
COP	×	×	×	×	Plantar pressure analysis system
Patellar circumference	×	×	×	×	Tape rule
SF-36	×	×	×	×	SF-36
Knee proprioception	×	×	×	×	CSMI
Knee ROM	×	×	×	×	Joint motion ruler
Pain	×	×	×	×	VAS score
Knee function	×	×	×	×	HSS score
Walking function	×	×	×	×	Holden classification

COP, centre of pressure; CSMI, Computerised Sports Medicine; HSS, Hospital for Special Surgery; ROM, range of motion; SF-36, 36-Item Short Form Health Survey; VAS, visual analogue scale.

Quadriceps and hamstring muscle strength

The quadriceps and hamstrings contract during distinct movement patterns. Muscle strength for both muscle groups will be assessed using a standardised MMT protocol. MMT is used to evaluate the knee joint muscle strength. It does not require any equipment when performing strength evaluation of subjects. It applies the following ratings:²² 0 as zero (O), 1 as trace (T), 2 as poor (P), 3 as fair (F), 4 as good (G) and 5 as normal (N). This procedure evaluates the function and strength of individual muscles based on the effective performance of a movement in relation to the forces of gravity and resistance. This method is simple, easy and has been widely used in clinics. For the quadriceps, participants will be seated with their knees flexed to 90 degrees. The examiner will stabilise the proximal thigh with one hand while applying gradual resistance against the anterior aspect of the distal tibia, instructing the participant to 'push forward against my hand while extending your knee'. To evaluate the hamstrings, the participant will maintain the seated position while the examiner stabilises the pelvis and applies progressive resistance to the posterior distal tibia, instructing them to 'pull backward against my hand while bending your knee'. All tests will be conducted bilaterally following this standardised procedure.

COP displacement parameters

COP displacement parameters will be measured using the Freemove platform (Sensor Medica, Rome, Italy) to assess postural stability. Participants will stand barefoot on the force platform with their feet positioned at shoulder width and their arms resting comfortably at their sides. Following a standardised protocol, three 30 s static standing trials will be recorded under eyes-open conditions, with 60 s rest intervals between trials to minimise fatigue.²³ The accompanying Freestep V.2.0 software (Sensor Medica, Rome, Italy) will calculate COP trajectory parameters, including total path length, sway velocity and the 95% confidence ellipse area. The software's symmetry index algorithm will also quantify weight-bearing distribution between the right and left lower extremities. All tests will be administered in a controlled laboratory setting by trained research staff adhering to standardised operating procedures.

Knee proprioception

Knee proprioception was evaluated using an isometric test and training system (Computerised Sports Medicine, CSMI) with the passive joint position replication method.²⁴ Patients wore loose-fitting athletic shorts to expose the affected limb and were seated in an adjustable chair with hips and knees flexed at 90°. Their thighs, ankles and lower backs were immobilised to prevent compensatory movements. Blindfolds and earplugs eliminated visual and auditory distractions. A computer programme selected proprioceptive measurement positions at 30° and 60°. For each test, the knee started at 90° flexion while the dynamometer arm moved at a constant

angular velocity of 2 rad/s. The angular error between the target and reproduced positions was recorded. Each position was tested three times in a quiet environment. The tester refrained from providing any cues to the patient.

SF-36 quality of life questionnaire scores

The SF-36 Health Survey, an internationally validated quality of life instrument, will be used to assess health-related quality of life across eight domains: physical functioning, role limitations due to physical health, bodily pain, general health perceptions, vitality, social functioning, role limitations due to emotional problems, and mental health. Its 36 items evaluate these health dimensions over a 4-week recall period. Participants will independently complete the questionnaire under supervised conditions at each assessment time point. Domain scores will be transformed to a 0–100 scale using standard scoring algorithms, where higher scores represent better health status and quality of life.²⁴

Holden walking function classification

The Holden Functional Classification assesses an individual's walking ability. It categorises this ability into distinct levels, allowing clinicians and rehabilitation therapists to objectively evaluate and monitor patient progress. The system comprises levels 0 through 5, with each level representing a different degree of independence and the amount of assistance required for walking.²⁵ A higher level corresponds to greater walking stability.

Other outcomes

Safety assessment

Throughout the trial, adverse events will be documented by patients, physicians and physiotherapists using a dedicated questionnaire. Physiotherapists and researchers will classify each event within 24 hours of its occurrence as either related or unrelated to the treatment. Treatment-related adverse events typically encompass infection, swelling or significant discomfort and pain. Subjects will receive appropriate medical care and compensation for any adverse reactions determined to be a direct consequence of the trial.

Patients' compliance assessment

Patient compliance will be monitored throughout the observation period. Detailed records will document any instances of loss to follow-up or extraction, along with the corresponding reasons.

Statistical analysis

SPSS V.25.0 statistical software was used to analyse the experimental data, using the χ^2 test to analyse the categorical variables. Normality of the data distribution was assessed with the Shapiro-Wilk test. The normal-distribution data were expressed as the mean $\pm$ SD ($\bar{X} \pm S$). independent samples t-test was used to test for significant differences between groups; repeated-measures and a one-way analysis of variance was used to measure the differences of all parameters at each time point, and

post hoc analyses were carried out using the Fisher's least significant difference method. The non-normally distributed data were given as medians and IQRs, and non-parametric tests were used to analyse the data. The significance level is established at 0.05 and the limits of the CI at 95%.

Data management and quality control

Data will be recorded on case report forms (CRFs). All study team members will undergo comprehensive training before the study commences, covering the study objectives, treatment protocols and quality control measures. All study documents—including informed consent forms, screening forms, CRFs and treatment records—will be maintained under strict confidentiality. Access to the data and its analytical use will be limited to the research team. A dedicated monitor will review CRFs and rehabilitation treatment records weekly to verify their accuracy and completeness. The principal investigator will convene regular meetings to identify, discuss and resolve any issues emerging during the observation period, thereby upholding the highest standards of data quality and integrity.

Ethics and dissemination

This study involves human participants and was approved by the Medical Ethics Committee of Nanjing Drum Tower Hospital, the Affiliated Hospital of Nanjing University Medical School (approval number: 2025-0141-01). The trial is registered at ClinicalTrials.gov (ID: chiCTR2500100566). Written informed consent will be obtained from all participants, and the findings will be disseminated through peer-reviewed publications and academic conferences.

DISCUSSION

This study investigated the therapeutic effects of MLB on functional recovery in patients with postoperative swelling following ACLR and patellar fracture. The hypothesis is grounded in prior research that confirmed the efficacy of MLB for functional recovery in patients with ACLR and early postoperative patellar fracture. By delivering sustained, moderate compression, MLB enhances lymphatic return while minimising adverse effects on the skin and circulation. This mechanism accelerates functional recovery and reduces swelling.²⁶

This study protocol integrates MLB with conventional rehabilitation for patients experiencing postoperative swelling following ACLR and patellar fracture. It investigates the effects of MLB application during the acute, subacute and chronic recovery phases. A 1-year follow-up will comprehensively evaluate the long-term impact of MLB on functional recovery after these surgical procedures.

From a mechanistic perspective, the compression from the elastic bandage elevates tissue hydrostatic pressure, which mitigates oedema caused by capillary leakage after

microvascular damage.^{27–29} This reduction in swelling helps prevent hypoxic injury to surrounding tissues.³⁰ MLB promotes the resolution of swelling and enhances joint stability, thereby facilitating early strength training. Previous studies have more frequently applied MLB to patients with acute ankle sprains. Those investigations demonstrated that MLB effectively reduced swelling and improved motor function in this patient population.³⁰ A RCT of 136 patients undergoing knee replacement found that MLB improved mobility while reducing swelling and pain at discharge. However, that study only assessed patients in the early postoperative period after TKA and did not include long-term follow-up.

This study is distinctive in its investigation of a programme design that integrates early MLB application with conventional rehabilitation while also planning for long-term intervention and follow-up observation. The approach is theoretically multifaceted. Long-term follow-up enables a systematic evaluation of the programme's sustained effects on functional recovery and the prevention of long-term complications. Nevertheless, limitations exist, including the potential for patient compliance and cooperation to decline over time, which may result in incomplete data or increased missed appointments.

In terms of clinical practice, if this study confirms the significant efficacy of MLB, it can be promoted as a standardised rehabilitation programme for patients with postoperative swelling after ACLR and patellar fracture. This is especially true for patients with severe swelling or poor adherence to traditional training. The 'early intervention+long term management' model proposed in this study transforms postoperative rehabilitation from traditional segmental management into continuous management of the entire process. This model promotes the establishment of an integrated rehabilitation system that includes preoperative assessment, intraoperative protection, postoperative early intervention and long-term follow-up. MLB is a simple, practical form of physical therapy that provides a feasible solution for standardised postoperative rehabilitation in primary care institutions.

In summary, this protocol explores the effects of MLB in the acute, subacute and chronic phases of the postoperative clinical application of ACLR and patellar fracture. It does so through a rigorous RCT design, especially by extending the time between return visits to comprehensively assess the long-term impact of MLB on the functional recovery of postoperative patients with ACLR and patellar fracture. Combining MLB with conventional rehabilitation targets postoperative swelling and may accelerate patients' return to normal life by optimising the rehabilitation pathway.³¹ Additionally, this study is expected to establish a scientific foundation for clinical decision-making, enhance the personalisation and precision of postoperative rehabilitation programmes, and improve overall treatment outcomes and patient quality of life.³²

Trial status

Recruitment for the trial started on January 1, 2025, and is expected to finish by January 1, 2027.

Contributors XH, as the principal investigator of the study, led all phases of the research, was responsible for developing the conceptual framework and conducting investigations, and participated in drafting and critically revising the manuscript to ensure its validity. WC, JW and HK were involved in the study design and participated in drafting and revising the manuscript. XR and CS contributed to manuscript revision, supervision, resource acquisition, project coordination and conceptual development. CS is the guarantor.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

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